

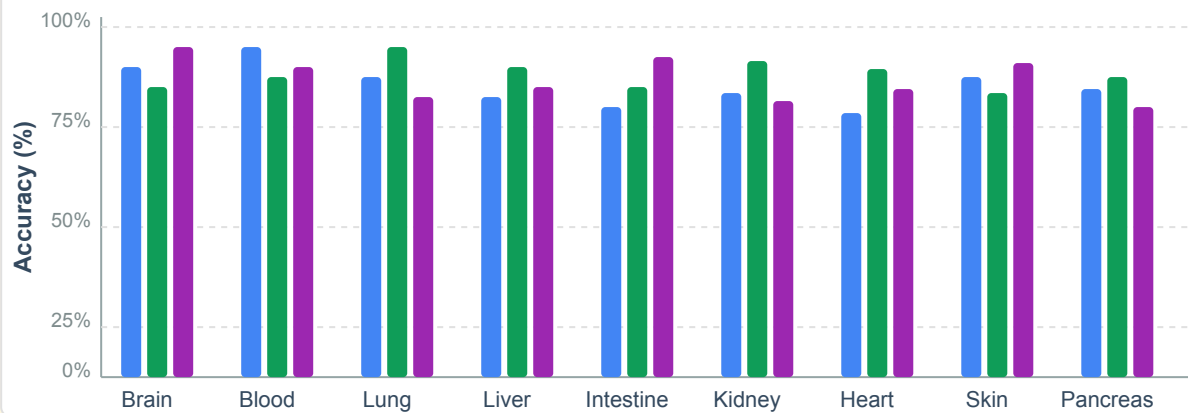
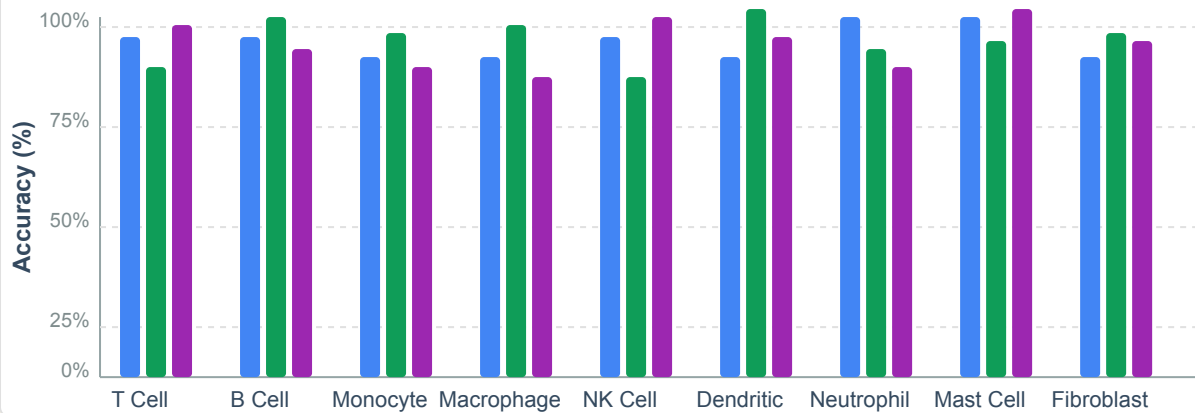
a

Complementary Strengths of Different LLMs in Cell Type Annotation

GPT

Claude

Gemini



b

Input Preparation SOP

Input: Clustered Data

- Pre-processed scRNA-seq data
- Clusters identified (e.g., Leiden)

Differential Expression Analysis

- Identify genes upregulated in each cluster vs. all others
- e.g., Wilcoxon rank-sum test
- Result: Top N marker genes per cluster

Pairwise DE Analysis (Optional)

- Identify genes distinguishing sibling clusters (sharing the same parent cluster)
- Result: Sister-specific marker genes

c

Framework Inputs

Tissue Information

- Tissue type: Human Lung
- Developmental stage
- Sample condition

Cluster Marker Genes

- Top N differentially expressed
- From Panel b: "Differential Expression Analysis"
- e.g., COL6A2, PCOLCE, CALD1

Sister Cluster Markers (Optional)

- Genes distinguishing sibling clusters (from Panel b)
- Used for resolving similar cell type annotations

e

Round 1: Initial Analysis & Discussion

GPT-4: "Key marker CD68 strongly indicates macrophage lineage"

Claude: "CD11c expression suggests dendritic cell identity"

Gemini: "MARCO expression patterns typical of specific macrophage subtypes"

Initial Evidence Analysis:

- Pan-Myeloid: CD68, CSF1R
- DC-associated: CD11c, CLEC9A
- Tissue-specific: MARCO, CD206

Initial Consensus Status:

- 4/6 LLMs: Macrophage (67%)
- 2/6 LLMs: Dendritic Cell (33%)

CC
Consensus
CheckerConsensus?
Yes
No (67%)

Round 2: Refinement & Evidence Review

Claude: "Tissue context important - lung sample suggests alveolar environment"

Gemini: "MARCO and CD206 co-expression specifically enriched in alveolar macrophages"

Qwen: "Pattern inconsistent with classical dendritic cells or interstitial macrophages"

Consensus Building:

- Tissue-niche specific markers present
- Alveolar environment considerations
- Ruled out other myeloid subtypes

Consensus Building:

- 5/6 LLMs lean toward Alveolar Macrophage
- Tissue context explains marker profile
- Strong evidence for specialized subtype

CC
Consensus
CheckerConsensus?
Yes
No (85%)

Round 3: Final Consensus & Validation

GPT-4: "All evidence supports alveolar macrophage annotation"

Claude: "CD11c expression typical in alveolar macrophages, not exclusive to DCs"

Gemini: "High confidence in alveolar macrophage identity with tissue context"

Final Evidence:

- Definitive alveolar macrophage signature
- Tissue-specific marker expression patterns

Final Status:

- Unanimous agreement on Alveolar Macrophage
- Strong tissue-specific context understanding

CC
Consensus
CheckerConsensus?
Yes (100%)
No

Final Output

Annotation Results

Cluster	Initial Predictions	Consensus	CP
C1	T cells (4/6) NK cells (2/6)	● T cells	0.67
C2	B cells (6/6)	● B cells	1.0
C3	Mac (4/6) DC (2/6)	● Alv. Mac	1.0

Initial Agreement ● Initial Dispute ●

Discussion Log

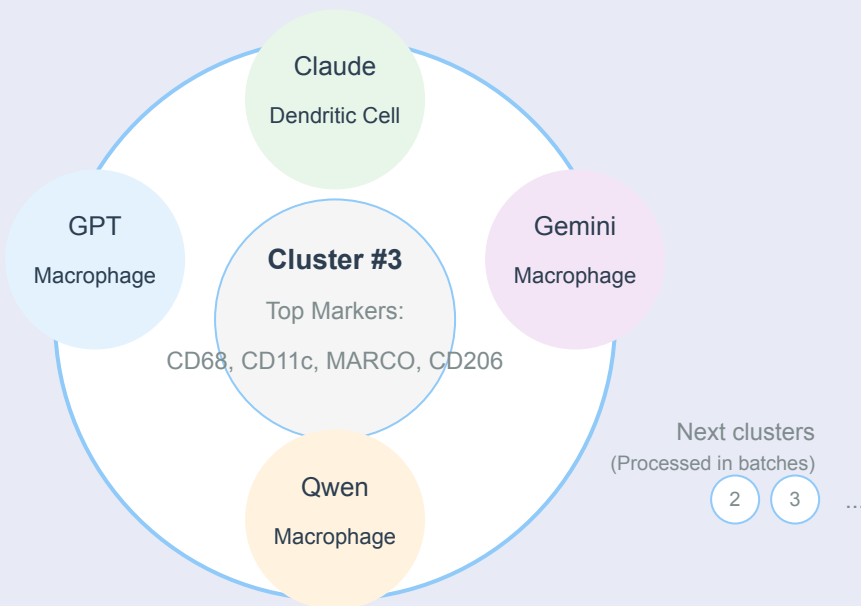
COL6A2 indicates myofibroblasts

CALD1, MYL6 are contractile markers

Consensus: Myofibroblast (ECM+, contractile+)
CP: 0.75

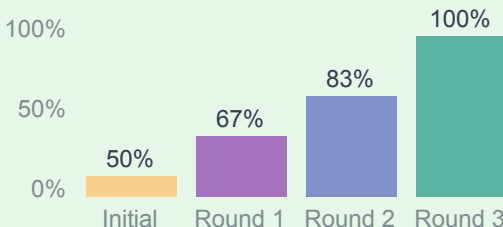
d

Initial Cluster Prediction



f

Consensus Evolution



Quality Assessment

- Full LLM agreement (6/6)
- Consensus Proportion (CP): 1.0
- Shannon Entropy (H): 0.0
- Marker consistency: High